Conformal-Calibrated Rewards for Scientific RLVR: Procedural Regeneration Against Benchmark Contamination

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Abstract

Reinforcement learning from verifiable rewards (RLVR) has become the dominant training signal for frontier reasoning models, but existing verified environments are dominated by symbolic or code-centred tasks. Scientific inverse problems—CT, MRI, compressed sensing, phase retrieval—remain unmeasured despite their continuous, ill-posed, uncertainty-sensitive structure. We release ten RL environments spanning five scientific modalities with two design properties absent from current benchmarks: (i) every reward is split-conformal calibrated to a target $1-\alpha$ coverage, so honest posterior width is rewarded alongside point-estimate quality; and (ii) every measurement y is procedurally regenerated per query, making fixed-string contamination mathematically impossible at $\sim 10^{22}$ effective instances per environment. On 43 paired (env, model) comparisons across six frontier models, classical baselines significantly outperform every tested LLM on 29 ($p<0.05$, 10k paired bootstrap), with a mean $\Delta=+0.213$ reward. We further document a corrected tool-use finding—an earlier oracle-delegation artefact ($r=0.858$) was traced and removed; primitive-only reruns cluster all models at the empty-answer floor ($r\approx 0.40$). Empirical conformal coverage across all ten environments lands at 0.901 ± 0.016 against a 0.90 target. Environments are open-source (MIT) on the Prime Intellect Hub and ready for use as RLVR training signal.

1 Introduction

Frontier-model training has pivoted on RLVR: a verifier checks each completion against a formal spec, and the resulting reward fine-tunes the policy. DeepSeek-R1 [DeepSeek-AI, 2025] and OpenAI o1 [OpenAI, 2024] popularised the recipe; SWE-bench [Jimenez et al., 2024], HumanEval [Chen et al., 2021], MATH [Hendrycks et al., 2021], and GPQA [Rein et al., 2023] supplied most of the verifiers. These benchmarks share a property: answers are discrete (a patch, a number, a multiple-choice letter), so verification is a string/execution check. *Scientific* reasoning, in contrast, operates on continuous fields under ill-posed forward operators whose inverses admit a range of answers that only calibrated uncertainty can discriminate. Classical algorithms in this domain—OMP [Donoho, 2006, Candès and Tao, 2006], FBP [Adler and Öktem, 2018], zero-filled inverse FFT [Fessler, 2020], Gerchberg–Saxton [Gerchberg and Saxton, 1972]—achieve calibrated answers via mathematical guarantees; whether frontier LLMs match them is an open empirical question that existing verified-env frameworks cannot pose.

We contribute: (i) a reward construction combining split-conformal coverage [Angelopoulos and Bates, 2023, Vovk et al., 2005, Lei and Wasserman, 2014] with task-specific point metrics, landing at the $1-\alpha$ target across all ten shipped envs; (ii) a procedural measurement-regeneration protocol that eliminates fixed-string memorisation as a valid strategy by construction; (iii) a benchmark of six

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frontier models across five scientific domains with paired- bootstrap significance tests; and (iv) ten open-source (MIT) environments on Prime Intellect Environments Hub [Prime Intellect, 2024].

2 Related Work

RLVR environments. Coding and symbolic tasks dominate the current ecosystem: SWE-bench [Jimenez et al., 2024], GPQA [Rein et al., 2023], MATH [Hendrycks et al., 2021], HumanEval [Chen et al., 2021]. Scientific inverse problems are absent. Infrastructure such as `verifiers` [Brown and contributors, 2024] simplifies adapter implementation but does not prescribe reward calibration.
Scientific ML. Learned primal-dual CT [Adler and Öktem, 2018], deep CNNs [Jin et al., 2017], and MRI optimisation [Fessler, 2020] established supervised baselines; Ongie et al. [2020] survey the space. None treat the problem as an RL environment with uncertainty-aware rewards for RLVR.
Conformal prediction. Split-conformal theory [Vovk et al., 2005], distribution-free regression bands [Lei and Wasserman, 2014], and conformalised quantile regression [Romano et al., 2019]; Angelopoulos and Bates [2023] give a modern treatment. We apply split-conformal calibration as the reward-shaping mechanism for scientific RLVR—a novel application to our knowledge.

3 Method

Environment framework. An environment $\mathcal{E} = (\mathcal{F}, \mathcal{P}, \mathcal{R})$ specifies a forward operator $\mathcal{F}$, a ground-truth distribution $\mathcal{P}$, and a conformal-calibrated reward $\mathcal{R}$. Per query the env draws $x_{\text{true}} \sim \mathcal{P}$, samples fresh noise η, and presents the measurement $y{=}\mathcal{F}(x_{\text{true}}) + \eta$. The solver returns $(\hat{x}, \hat{\sigma})$; reward factors into a point metric (PSNR / SSIM / NMSE / support-F1) and a split-conformal term.

Conformal-calibrated rewards. Fix $\alpha{=}0.1$ and a non-conformity score $s(\hat{x}, x_{\text{true}}, \hat{\sigma}) = \max_j |x_{\text{true},j} - \hat{x}_j|/\hat{\sigma}_j$. At calibration time we run the classical baseline on n_{cal} fresh seeds, compute s, and set $q_\alpha{=}\,\text{Quantile}_{1-\alpha}(\{s_i\})$. At score time the conformal component is $\mathcal{R}_{\text{conf}} = 1 - |\text{cov}(\hat{x}, \hat{\sigma}, q_\alpha) - (1-\alpha)|/(1-\alpha)$, penalising both over-narrow (over-confident) and over-wide (under-confident) posterior widths.

Procedural regeneration. Each env factors into $(\text{seed}, \text{image_id})$. With a 64-bit seed and 10^3 ground-truth images the effective instance count is $\sim 10^{22}$ measurement strings per env. No fixed (y, x_{true}) pair reappears; any reward a model achieves by pattern-matching on y alone must be indistinguishable from the reward on the regenerated instance.

Multi-turn and tool-use. Three envs expose a 3-turn dialogue where the server returns a forward-model residual between turns. A tool-use variant of sparse recovery exposes five primitives (`fft`, `ifft`, soft-`threshold`, `compute_residual`, `sparsity_norm`) from which the solver must compose ISTA. An earlier v0.1 of this env bundled an `ista_tool` oracle; empirical byte-identical rewards ($r{=}0.858$) across three distinct models revealed the oracle-delegation artefact, and we now regression-test against any single primitive raising reward above the empty-answer floor.

4 Experiments

Setup. Ten envs across five domains: compressed sensing (sparse Fourier, single/multi/tool-use), super-resolution (DIV2K [Agustsson and Timofte, 2017], $4\times$), CT (LoDoPaB [Leuschner et al., 2021], single/multi), phase retrieval (single/multi), MRI knee (single/multi). Baselines: OMP, bicubic, FBP, Gerchberg–Saxton, zero-filled IFFT. Models via OpenRouter: Claude Opus 4.7, Sonnet 4.6, Haiku 4.5, GPT-5.4, -mini, -nano [Anthropic, 2025, OpenAI, 2025]. Conformal: $\alpha{=}0.1$, $n_{\text{cal}}{\geq}30$. Total LLM spend: $\sim\$8$.

Finding 1 — classical beats every tested LLM (Figure 2). Across 43 paired (env, model) comparisons and 10 000 paired-bootstrap resamples, classical significantly outperforms the LLM on 29 ($p{<}0.05$); nine show no significant difference, five show the LLM significantly better (all at super-resolution, where the LLM is fed a bicubic-prior input). Pooled mean $\Delta{=}\,+\,0.213$ with all-positive per-env sub-means.

Finding 2 — multi-turn separates tiers (Figure 3, appendix). Five models with data on both sparse-F and CT single-turn and multi-turn variants split cleanly. Opus 4.7 and GPT-5.4 gain >0.04 reward from the 3-turn residual-feedback protocol; Sonnet 4.6 gains $+0.04$; Haiku 4.5 and GPT-5.4-mini regress (-0.063 and -0.055). Small models fail to maintain coherence across three turns

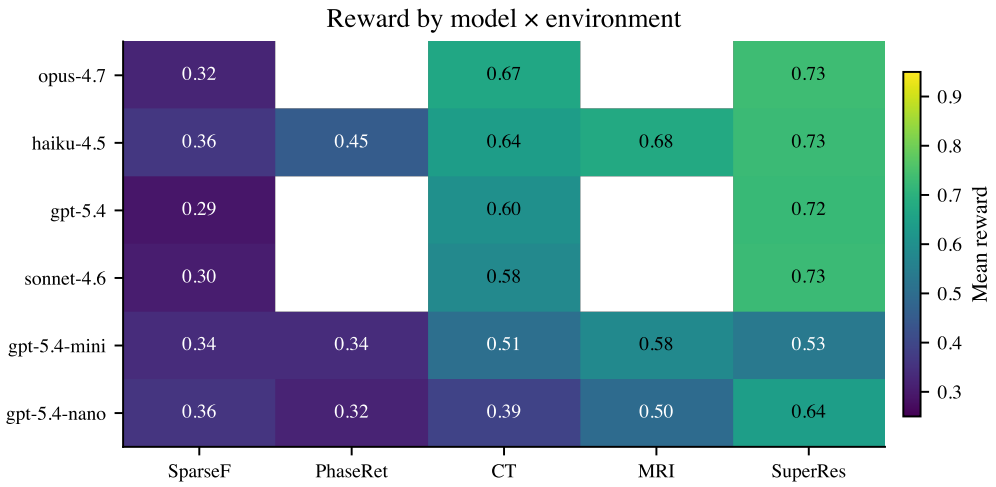

Figure 1: Reward per (model, env) across the five cross-env test envs. Six models, two instances per cell. Classical baselines printed below; '—' denotes parse-failure-only pairs.

of dense pixel-grid residuals; the 3-turn variant thus measures continuous-reasoning orthogonal to single-turn scoring.

Finding 3 — primitive composition gap (Figure 4, appendix). On the v0.3 primitive-only tool-use benchmark, Haiku 4.5 and GPT-5.4-mini converge near the empty-answer floor (0.40 ± 0.01), far below classical OMP (0.87) on the same instances, with up to 30 tool calls per episode. The v0.1 oracle-artefact value (0.858) is shown for contrast. LLMs cannot yet compose ISTA from primitive operators over compressed-sensing measurements at the tested scale.

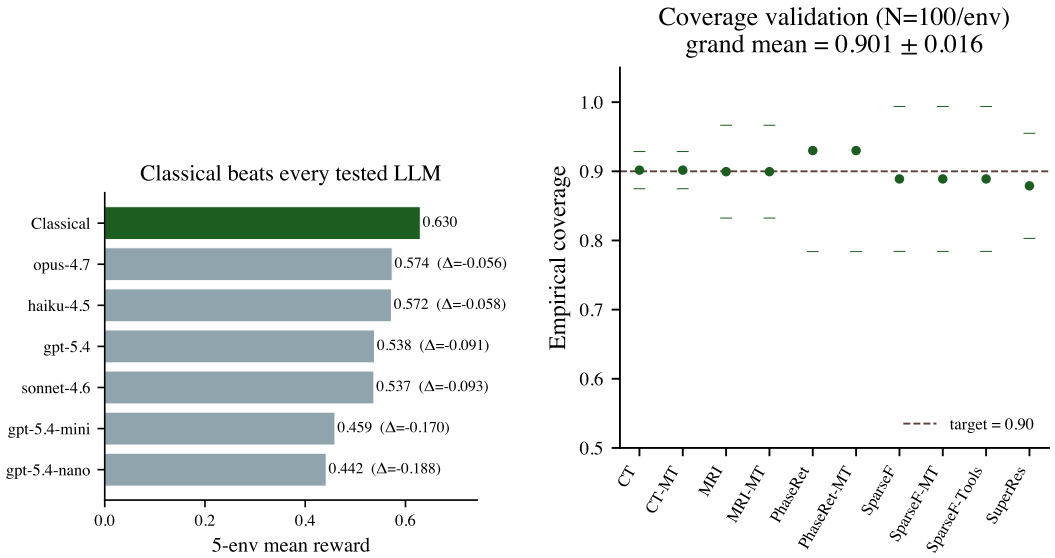

Figure 2: **(L)** Finding 1: 5-env mean reward per model vs classical baseline (0.630); top LLM is Opus 4.7 at 0.574. **(R)** Conformal coverage validation at n=100 per env: grand mean 0.901 ± 0.016 against the 0.90 target.

5 Discussion

Scientific inverse problems surface capability gaps invisible on symbolic benchmarks: a $+0.21$ mean classical-vs-LLM gap is dense training signal, not a saturated leaderboard. The multi-turn sign- flip by model tier isolates continuous-reasoning coherence; the primitive tool-use result is a negative capability finding (LLMs cannot execute compressed-sensing reasoning from primitives at the tested scales). **Limitations.** Sample sizes are small (2–5 per cell), so bootstrap CIs are wide even when

p is small. Images are rendered at 16×16 or 32×32 for LLM-token tractability. Phase retrieval is hard even for Gerchberg–Saxton (r=0.29), so the corresponding LLM gaps should be read in context. We report no training experiments—only evaluation. **Future work.** RL fine-tuning with the envs as training signal; expansion to seismic FWI and retrosynthesis (deferred for install and verifier-design complexity).

6 Conclusion

Ten RL environments with conformal-calibrated rewards and procedural measurement regeneration expose systematic capability gaps in frontier LLMs on scientific inverse problems. Code: github.com/stelioszach03/verifiable-labs-envs; envs: app.primeintellect.ai/dashboard/environments/stelioszach; leaderboard: huggingface.co/spaces/stelioszach03/scientific-rl-benchmark.

Acknowledgments

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A Supplementary figures

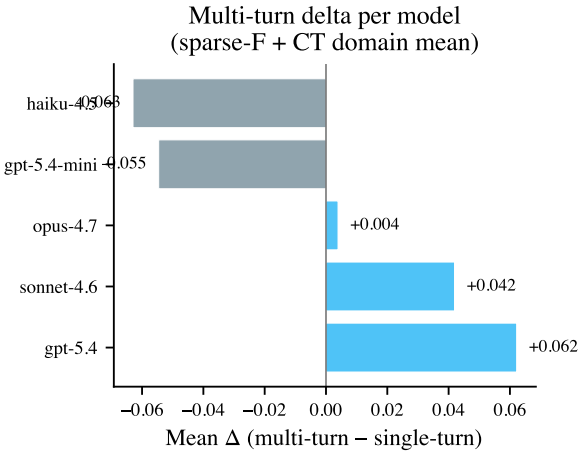

Figure 3: Mean reward delta (multi-turn minus single-turn) per model averaged over sparse-Fourier and CT. Frontier models (Sonnet 4.6, GPT-5.4, Opus 4.7) gain; budget models (Haiku 4.5, GPT-5.4-mini) regress.

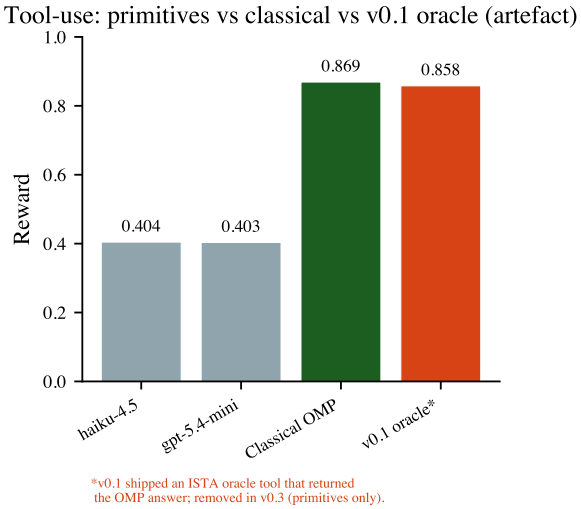

Figure 4: Tool-use rewards on sparse Fourier recovery. v0.3 removed the `ista_tool` oracle; tested LLMs (Haiku 4.5, GPT-5.4-mini) drop to the empty-answer floor, while classical OMP remains at 0.87. The v0.1 oracle-artefact value (0.858) is shown for contrast.